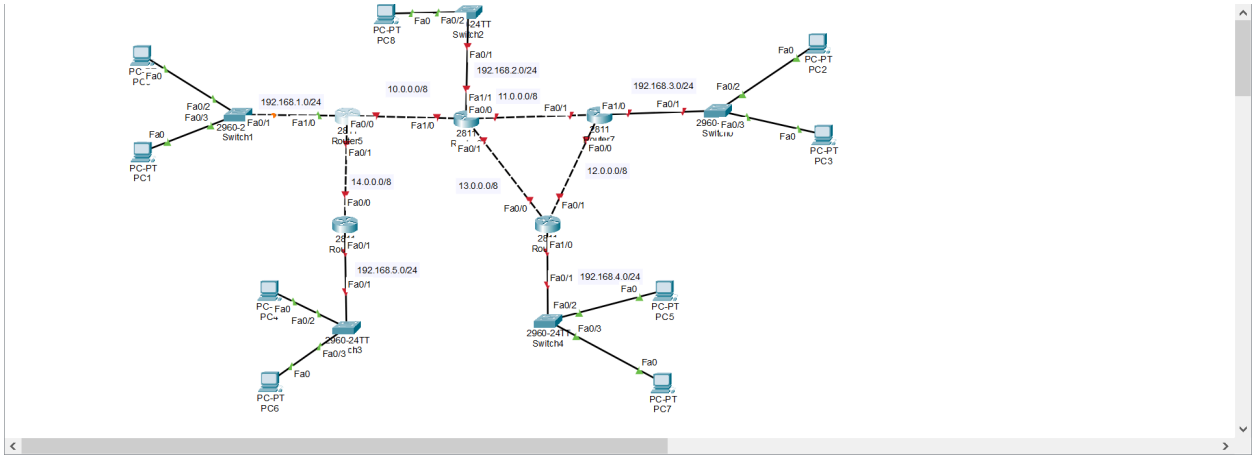
NETWORK DESIGN

Overview:

Static routing was implemented to enable full connectivity between all 5 LAN segments(192.168.1.0- 192.168.5.0) across 5 routers. Each router was configured with routes to all remote networks that are not directly connected.

Subnet mask standard:

1) All 192.168.x.0 networks=255.255.255.0(/24)
2) All 10.0.0.0-14.0.0.0 = 255.0.0.0(/8)

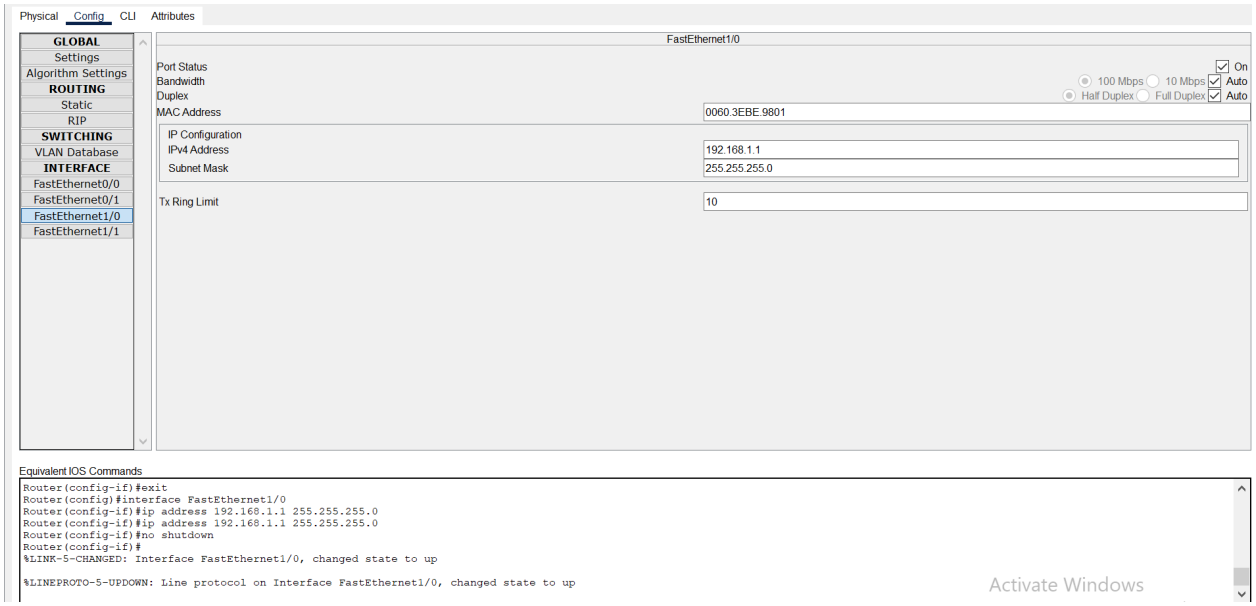


Devices used:

Device Type	Model	Quantity	purpose
Router	2811	5	Interconnect LANs
Switches	2960	5	Access layer
PCs	PC-PT	9	End devices
Cables	Straight + DCE	20	LAN +WAN links

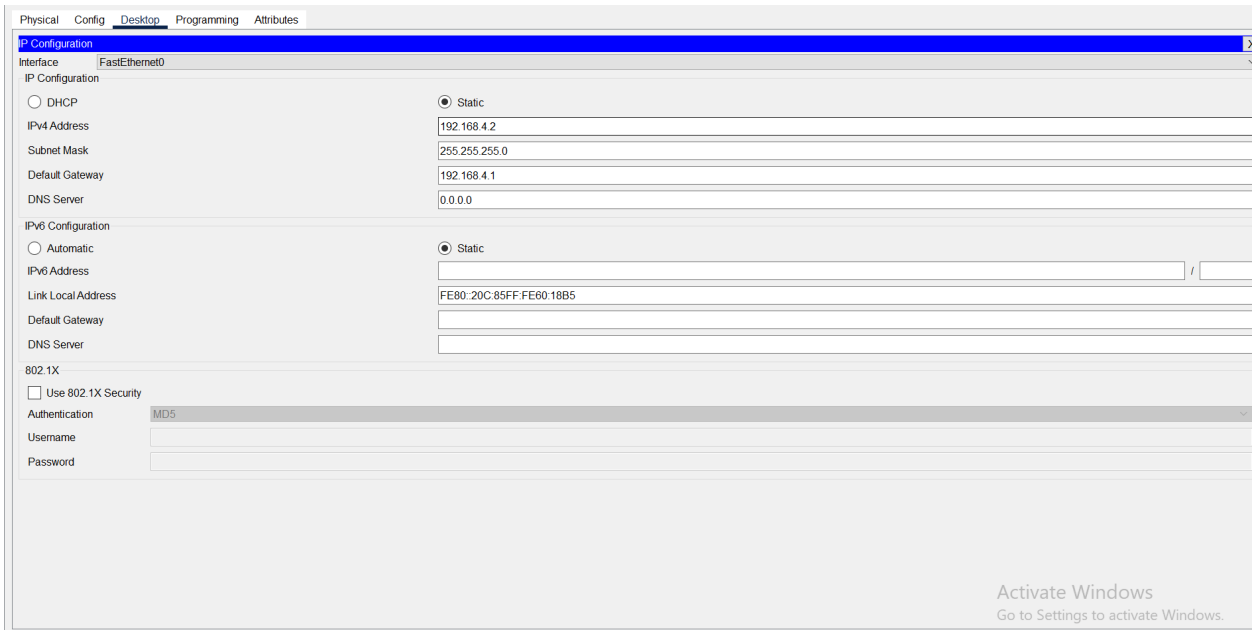
Assigning IP addresses to router interfaces:

IP addresses we assigned to routers interfaces using the graphical config tab in Packet Tracer. Each interface was selected, its IP addresses and subnet mask were entered, and the ports were turned on. This method automatically enables the interfaces.



Assigning IP addresses to PCs:

IP addresses for end devices were assigned manually via Desktop > Configuration. Each PC was given a unique IP address within its LAN, with the routers FastEthernet interface as the default gateway.



Static Routing Configuration:

Static routes were configured on all routers to enable communication between remote networks. Each routes defines the destination networks, the subnet mask and the next-hop IP addresses that lead to that destination.

The command format used is ip route
[network] [mask] [next-hop].

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 11.0.0.0 255.0.0.0 10.0.0.2
Router(config)#ip route 12.0.0.0 255.0.0.0 10.0.0.2
Router(config)#ip route 13.0.0.0 255.0.0.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

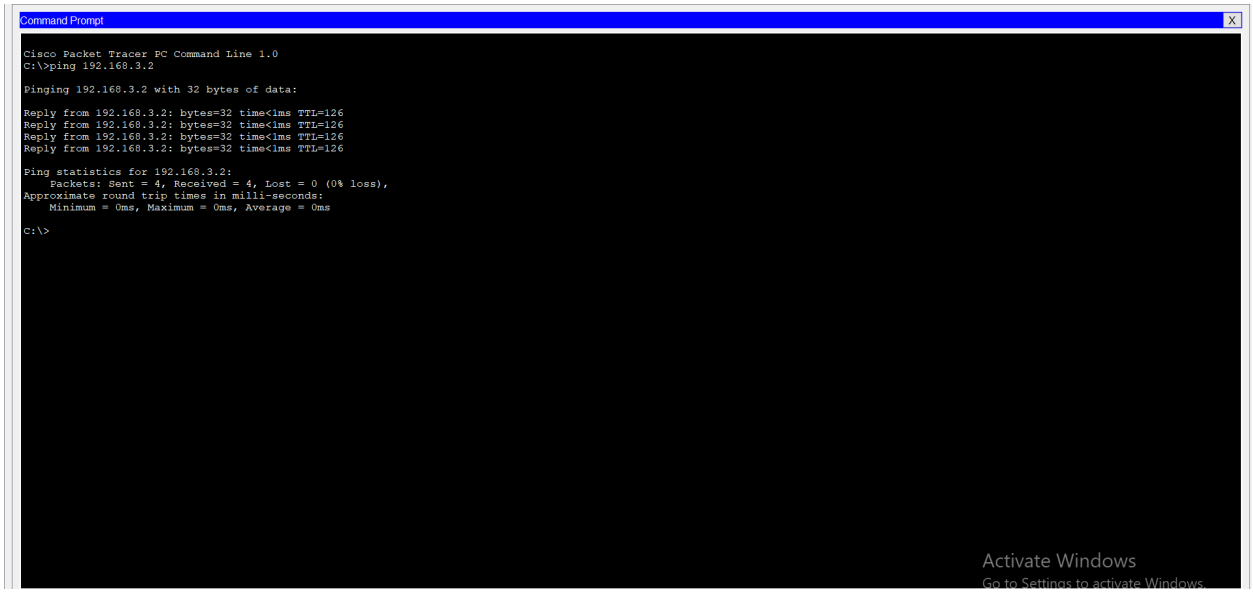


Router con0 is now available
```

Testing and verification:

After configuration:

1) Run do wr on all routers to ensure all configuration is saved.
2) All interfaces are on .
3) Test connectivity.



Explanation for your report:

IP route [destination_network] [subnet mask] [next_hop_ip] command tells the router “ to reach this destination, send packets to this next-hop address”. Static routing was chosen for this project because it gives full control over the paths and is secure for a small enterprise network.